[INCH-POUND]
A-A-59252
June 16, 1998
SUPERSEDING
MIL-E-3118C
12 February 1974

COMMERCIAL ITEM DESCRIPTION

ETCHERS, ELECTRIC, PORTABLE, 120-VOLT A.C.

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers electric etchers of electric arc and vibrating point type.

2. CLASSIFICATION. Electric etchers covered by this CID are of the following types as specified (see 7.2).

Type I - Light duty etcher, 500 watts.
Type II - Heavy duty etcher, 1,000 watts.
Type III - Heavy duty etcher, vibrating type.

3. SALIENT CHARACTERISTICS.

3.1 Description. Type I and II etchers shall consist of a case, transformer, control panel, power "on-off" switch, power cable with plug, etching tool lead, etching tool or stylus, ground clamp, and a ground plate when furnished. Type III etchers shall consist of a drive unit, reciprocating shaft, collet or chuck, switch, power cable, and replaceable etching points. Etchers shall operate on 120 volts (rated voltage), 50 to 60 Hertz, alternating current (AC).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Officer In Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, CA 93043-4410, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5130

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3.2 Standard commercial product. The etchers shall, as a minimum be in accordance with this CID and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this CID but which are a part of the manufacturer's standard commercial product, shall be included in the etchers being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Design. The etcher shall be designed to etch on metals by the electric-arc process for types I and II and vibratory-scarifying process for type III.

3.3.1 Insulation resistance. Insulation of the etcher shall be dielectric.

3.3.2 Plugs. Plugs on etcher power cables shall be rated at not less than 15 amperes, 125 volts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means of preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or permanently deformed.

3.3.4 Power cable. The power cable for etchers shall be not less than 5-foot (1 524 millimetre (mm)) length of commercial type, three-conductor cable of suitable size. The cable shall be equipped with the plug specified in 3.3.2. The length of cable shall be measured from the point of entry into the tool.

3.3.5 Types I and II.

3.3.5.1 Heat output. Type I and II etchers shall be capable of operating for 30 minutes at heat output of 500 to 1,000 watts respectively in "on" and "off" duty cycles. The transformer temperature shall be not greater than 149 degrees Fahrenheit (°F) (65 degrees Celsius (°C)) above ambient. The etching point shall withstand the heat developed without distortion or blunting of the tip. Type I shall have not less than 3 heat-taps. Type II shall have not less than 10 heat-taps with not less than three taps below 300 watts.

3.3.5.2 Weight. The weight of type I and II etchers shall be not greater than 25 and 35 pounds (11.34 and 15.88 kilogram (kg)), respectively.

3.3.5.3 Case. The case of type I and II etchers shall be made of sheet steel. The base of the case shall have five embossed dimples of not less than 0.375-inch (9.53 mm) spherical radius that shall raise the inside base of the etcher case. Alternatively, rubber pads of not less than 0.875-inch (22.23 mm) diameter by 0.375-inch (9.53 mm) thick may be substituted for the dimples. The pads shall be attached with locked screws. The case shall be equipped with a handle with a hand grip. Space shall be provided within the case for stowage of the power cable, etching tool with spare points, etching point and ground leads, and a ground plate when furnished. Alternatively, an integral part of the case, suitable for use as a ground for the etching tool, may be substituted for the ground plate (see 3.3.5.8).

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3.3.5.4 Transformer. The transformer shall be designed for the duty specified herein on a rated input voltage of not less than 120-volts AC, without creating a fire hazard. The transformer shall be mounted rigidly and securely in the etcher case so that the transformer shall not become loosened or detached from its mounting. The use of an adhesive for mounting shall not be acceptable.

3.3.5.5 Control panel. The control panel shall have an indicator light to indicate when power is on, two output terminal studs, an “on” and “off” power switch, and a rotary heat-tap selector with a pointer and legible dial for indicating the applicable heat-taps (see 3.3.5.1). The output terminals shall be insulated to prevent any grounding through the hole cross sectional and panel surface areas.

3.3.5.6 Etching tool and ground leads. Etching tool and ground leads shall be not less than 3-foot (914.40 mm) length of suitable size of insulated wire cable. Each lead shall be equipped with a spade-type copper terminal at one end for connection to the transformer terminal studs. An etching tool shall be attached to the other end of the etching tool lead and a positive-type grounding copper terminal shall be attached to the other end of the ground lead.

3.3.5.7 Etching tool. The etching tool or stylus shall have an insulated holder of suitable size and shape for ease of handling and a replaceable etching point. The tool shall be provided with a spare etching point.

3.3.5.8 Ground plate. The ground plate of type I and II etchers, when furnished, shall provide electrical conductivity and have not less than a 25 square inch (16 129 square millimetre) surface area.

3.3.5.9 Ground clamp. The ground clamp of type I and II etchers, when furnished, shall provide electrical conductivity for etching large parts that cannot be conveniently placed on the ground plate.

3.3.6 Type III. The type III etchers shall be capable of operating continuously for one-hour so that the etching points shall withstand shock (see 3.3.6.6) without fracturing, bending or twisting.

3.3.6.1 Weight. The weight of type III etcher shall be not greater than 1-pound (0.45 kg).

3.3.6.2 Drive unit. The drive unit shall consist of a heavy duty vibrating unit. The housing shall be sized and shaped to provide a comfortable hand-grip and shall have a power switch. The etcher shall operate to provide the specified speed for the reciprocating shaft (see 3.3.6.3).

3.3.6.3 Reciprocating shaft. The reciprocating shaft shall be mounted in a heavy duty sealed bearing. The shaft shall reciprocate at not less than 7,200 strokes per minute.

3.3.6.4 Chuck. A collet or chuck shall be located on the reciprocating shaft and shall accommodate replaceable etching points.

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3.3.6.5 Switch. The “on-off” switch on the motor housing shall be placed in a convenient location for the operator. The switch shall be toggle, slide, or rocker-type, and shall be installed as part of the housing configuration.

3.3.6.6 Etching points. The etching points for type III etchers shall be wear- and shock-resistant. The etcher shall be furnished with six alloy steel etching points consisting of two with diamond tips and four with tungsten carbide tips.

3.4 Finish. Etchers shall have enameled or plated finishes on metal surfaces in accordance with manufacturer’s standard commercial practice, except in no case shall cadmium plating be used.

3.5 Identification plate. A corrosion-resistant metal or plastic identification plate, permanently and legibly marked with the following information, shall be securely attached to the etcher:

Manufacturers name.
Manufacturer’s model or serial number.
120 volts.
50-60 Hertz, AC only.

3.6 Fungus. When specified (see 7.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a standard commercially accepted varnish with the following exceptions:

a. Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed, need not be treated.

b. Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

4. REGULATORY REQUIREMENTS.

4.1 Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this commercial item description are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description.

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of ASTM SI-10 (IEEE/ASTM SI-10), and all other requirements of this commercial item description including

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form, fit and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 The Federal Acquisition Regulation (FAR) is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 ASTM Standard is available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

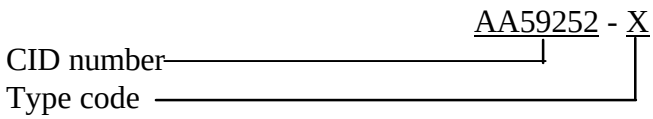
7.1.3 IEEE Standard is available from the Institute of Electrical and Electronics Engineers, IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Type required (see 2.).
c. When fungus protection is required (see 3.6).

7.3 Supersession data. This CID replaces Military Specification MIL-E-3118C, dated 12 February 1974.

7.4 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for items acquired to this description are created as follows:



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7.5 Metric units. The values stated in either inch-pound units or SI units are to be regarded separately as standard. Within the text, the SI units are shown in brackets. The values stated in each system are not exact equivalents; therefore, each system should be used independently of the other. Combining values from the two systems may result in non-conformance with the standard.

7.6 Subject term (key word) listing.

Engrave
Identification
Inscribe
Marking
Stylus

7.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - YD1
Air Force - 99

Review Activities:
Navy - AS, SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

(Project 5130-0400)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59252	**2. DOCUMENT DATE (YYMMDD)** 980616

3. DOCUMENT TITLE

ETCHERS, ELECTRIC, PORTABLE, 120-VOLT A.C.

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME KENNETH A. POLLOCK	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5612 (2) AUTOVON 551-5612
c. ADDRESS *(Include Zip Code)* OFFICER IN CHARGE, SEABEE LOGISTICS CENTER CODE 15E2, 4111 SAN PEDRO ST. PORT HUENEME, CA 93043-4410	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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